Reg. No. | | | | | | | | | | | [14ME14/24]

FIRST SEMESTER B.TECH. EXAMINATION, DECEMBER - 2015

MECHANICAL ENGINEERING SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions : Answer five full questions, choosing at least TWO from each part.

PART-A

1.	**a.**	Differentiate between renewable and non-renewable sources of energy with examples.	**06**
	b.	Explain with a new sketch solar flat plate collector	**08**
2.	**a.**	Explain the process of formation of steam with the help of suitable diagrams.	**08**
	b.	Distinguish between water tube and fire tube boiler.	**06**
3.	**a.**	Define compounding. Explain Pressure –Velocity compounding.	**09**
	b.	Write a note on principle of centrifugal air compressor.	**05**
4.	**a.**	Define the following.	**05**
		i. COP	
		ii. Ton of refrigeration	
		iii. Refrigeration effect	
		iv. Refrigerant	
		v. Relative COP	
	b.	Explain with a neat sketch vapor absorption refrigeration system.	**09**

PART-B

5.	**a.**	Explain with a neat sketch 4-stroke diesel engine.	**08**
	b.	The following observations refer to a trial on a single cylinder diesel engine.	**06**

Brake Power = 75KW;

Brake Thermal efficiency = 35%

Mechanical efficiency = 90%

CV of Oil = 40000 KJ/Kg

Determine

i. Indicated power

ii. Friction power and

iii. Mass of fuel used.

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6. a. Explain with a neat sketch radial drilling machine. **09**

b. With a neat sketch explain **05**

i. Knurling

ii. Facing

7. a. Classify the gear drives. Explain briefly the compound gear train. **07**

b. In a compound gear train of wheel A,B,C and D have 15, 30, 20 and 40 teeth respectively. The wheels B and C are keyed to the same spindle. If the wheel A runs at 400 rpm. Find the speed of wheel D. Sketch the arrangement if B meshes with A and C meshes with D. **07**

8. a. With a neat sketch explain the process of arc welding. **08**

b. Distinguish between soldering and brazing. **06**